



Stained Glass Panels

Hackathon Grasshopper Users Meeting

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The Problem

We want unique and beautiful elements in our spaces.

Problem: Custom ornament is expensive because it requires intense manual design and specialized craftsmanship.

Result: Budget constraints force us back into sterile, mass-produced minimalism.

Our Solution

We move the complexity from manual labor to software.

Platform: A configurator that lets anyone turn an image into a stained glass panel.

Output: Instant, 3D-printable files. No specialized craft skills required; no fortune needed.

The Inputs

Raster

Number of colours

Colours

Pattern of the panels

The Outputs

Visualisation

Model 3D in .glb

The workflow

Parallel tracks from the start. We split the team into:

- image-processing - Somia, Mateusz
- geometry - Martyna, Ewa
- web/render/deploy - Bartłomiej
- UI and UX design - Martyna

The Tools

Rhino 7 + Grasshopper

First geometry engine — image sampling, colour quantization, panelization, BREP frames/extrusion

Python / Jupyter

Early model transport from Grasshopper to the web viewer

Rust + wasm-bindgen

The production engine, compiled to WebAssembly (sg-wasm/)

three.js (WebGPU)

The 3D "beauty" renderer

Cloudflare + Wrangler

Hosting (no-build static site)

Illustrator/Photoshop

Logo (star rosette), grasshopper mascot, loading GIF, app mockups

Live Demo



STAINED GLASS

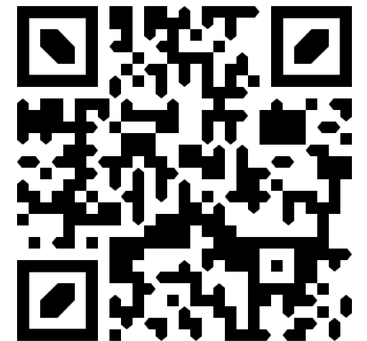
Pick your colours. Pick your pattern. Let the light do the rest.

Stained glass used to be the language of cathedrals and grand estates.
We're bringing it back — to your home, your studio, your space. Design
your own piece, choose your colours and pattern, and we'll print it. No
craft skills needed. No fortune required.

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What's Next?

Our prototype can develop in multiple ways, we propose this roadmap to advance our Idea:

- 1) Create viable way of connecting panels to each other using Printable frame exported using website
- 2) Line recognition of pictures uploaded on website
- 3) Optimised technical view with detail of connections and 2d drawings
- 4) The printer selection
- 5) Estimated price of the finished product
- 6) Gallery of the pictures with finished model
- 7) Online shop for special orders and infrastructure
- 8) 3D printer farms for orders
- 9) Metal printing of the mesh

Thank you!